
Where Molecular Energy Hierarchies Break: Chemically Resolved and Calibrated Uncertainty in HIP-NN

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Abstract

Hierarchical energy models expose internal corrections that ought to decay, but a correlated hierarchy score is neither a chemical explanation nor a calibrated error forecast. We train three strict-protocol HIP-NN replicas on QM9 molecules with at most eight heavy atoms and sequester every nine-heavy-atom molecule. The replicas attain 0.318 ± 0.011 kcal mol⁻¹ validation MAE and 0.789 ± 0.015 kcal mol⁻¹ on the molecular-size shift. HiBreak, a residual-independent and multiplicity-controlled chemical screen, freezes 6 motifs; on an untouched final half the first two attribute $83.6 \pm 0.2\%$ of errors above 1 kcal mol⁻¹ while matching 71.1% of molecules. HiPerCal transforms the complete three-order hierarchy profile into absolute-error units. Its 0.016 ± 0.003 kcal mol⁻¹ expected calibration error is slightly higher than the 0.013 ± 0.003 obtained from the original non-hierarchicality score alone, but its calibration-selected flags improve precision and recall to $49.9 \pm 1.1\%$ and $65.9 \pm 1.1\%$, respectively, from $38.5 \pm 0.6\%$ and $56.9 \pm 3.1\%$. A formula-stratified ANI-1x test then separates chemistry that transfers to off-equilibrium configurations from motifs specific to minimized QM9 geometries. These results turn an internal regularizer into a falsifiable applicability-domain diagnosis, while the reported misses show why it cannot serve as a safety certificate by itself.

1 Introduction

An average error is an incomplete safety case for a molecular energy model. The original hierarchically interacting particle neural network (HIP-NN) reached an atomization-energy mean absolute error (MAE) of 0.256 ± 0.003 kcal mol⁻¹ on a random QM9 split, yet $2.26 \pm 0.07\%$ of its predictions missed by more than 1 kcal mol⁻¹ [Lubbers et al., 2018, Tables I and II]. A rare error can determine the wrong conformer, screening rank, or trajectory when a model is deployed repeatedly. The practical question is therefore not only whether HIP-NN is accurate on average, but whether its own calculation exposes when and why it is unreliable.

HIP-NN is unusually well suited to that question. It writes the prediction as $\hat{E} = \sum_i \sum_{n=0}^2 \hat{E}_i^n$, where successive interaction orders expand the atomic receptive field. The non-hierarchicality

$$R(\mathcal{C}) = \sum_i \sum_{n=1}^2 \frac{(\hat{E}_i^n)^2}{(\hat{E}_i^n)^2 + (\hat{E}_i^{n-1})^2} \quad (1)$$

is large when higher-order corrections fail to decay. Lubbers, Smith, and Barros showed that error quantiles generally rise with R , but stopped short of asking which chemistry produces that rise or whether R has the units and calibration needed for a decision threshold [Lubbers et al., 2018]. A correlated score is not yet a predicted error: it may be mis-scaled, driven by a

diffuse set of unrelated structures, or confidently wrong in exactly the tail where intervention matters.

We study this gap under an explicit molecular-size shift. The energy model is trained only on pruned QM9 molecules containing at most eight heavy atoms and is evaluated on all molecules with nine heavy atoms. The architecture, energy loss, regularizers, optimizer, and early-stopping controller are held to the 2018 protocol. This choice turns a familiar benchmark into an applicability- domain experiment while preserving an external, paper-as-reported accuracy anchor. It also prevents architecture improvements from being mistaken for a better diagnosis of the original hierarchy.

Our first contribution is HiBreak, a chemically named failure library selected from sixteen graph and geometry hypotheses. Candidate definitions are independent of model residuals. A hash-defined discovery quarter screens them jointly for larger error and larger R , with false-discovery-rate control; an untouched final half supplies all confirmatory effects and the motif-to-failure Pareto curve. This design lets the library use residuals for discovery without reusing those molecules as evidence. Per-order atomic energies connect each selected motif to the particular decay step that weakens.

Our second contribution is HiPerCal (Hierarchy-Profile Calibration), a post-hoc map from six already available hierarchy and replicate-disagreement signals to absolute error. Signal directions are constrained to be nonnegative, and a monotone isotonic map supplies error units. Coefficients, bin edges, and the > 1 kcal mol⁻¹ operating point are learned on a calibration quarter and evaluated once on the same untouched final half used for confirmatory chemistry. We report every reliability bin, expected calibration error (ECE), precision, recall, false-positive rate, and false-negative rate, including individual false alarms and missed failures.

Finally, we test whether the equilibrium-geometry motifs survive a second kind of shift. Before model inference, we freeze a formula-stratified random sample of off-equilibrium ANI-1x structures [Smith et al., 2020]. Disjoint within-formula halves remove the otherwise dominant offset between QM9 and ANI-1x electronic-structure levels, leaving a relative-energy test across conformations and isomers. Taken together, these experiments distinguish three claims that are often conflated: a hierarchy can rank errors, its failures can have a compact chemical explanation, and its score can be calibrated into a useful decision quantity.

2 Study design

2.1 Data, target, and leakage barriers

QM9 contains DFT properties and optimized geometries for 133,885 neutral HCNOF molecules derived from GDB-17 [Ruddigkeit et al., 2012, Ramakrishnan et al., 2014]. We streamed the supplied archive and computed the 0 K atomization target as

$$E_{\text{at}} = \left(U_0^{\text{mol}} - \sum_{a \in \mathcal{C}} U_{0, Z_a}^{\text{atom}} \right) (627.5094740631 \text{ kcal mol}^{-1} \text{ Hartree}^{-1}), \quad (2)$$

reading both molecular and atomic reference values from the supplied QM9 files. We removed the supplied 3,054 uncharacterized identifiers and the eleven difficult-minimization identifiers. Three difficult identifiers already occur in the first list, so the set union excludes 3,062 unique molecules and retains 130,823. This overlap is recorded rather than double-counted.

Heavy atoms mean C, N, O, or F. The cleaned corpus contains 21,799 molecules with at most eight heavy atoms and 109,024 with exactly nine. For each seed, a new random permutation assigns 1,000 small molecules to early-stopping validation and the other 20,799 to training. No nine-heavy-atom index enters an optimizer batch, validation batch, or stopping decision. The three formal seeds are 117, 223, and 337.

The nine-heavy-atom identifiers are independently assigned by $u = 2^{-64} \text{uint64}[\text{SHA256}(20260731:\text{QM9_ID})]$, interpreting the first eight digest bytes as a big-endian unsigned integer, to discovery ($u < 0.25$, 27,196 molecules), calibration

($0.25 \leq u < 0.50$, 27,328), or final evaluation ($u \geq 0.50$, 54,500). Motif selection uses discovery only. HiPerCal fitting and threshold choice use calibration only. Accuracy and assignment files are computed for the full OOD set, but confirmatory motif and calibration claims use final evaluation only. The split rule and both model definitions were written to disk before a checkpoint was evaluated.

2.2 Paper-parity HIP-NN

The supplied `hippynn` implementation provides the scalar `Hipnn` interaction network and `HEnergy` hierarchical readout. We use these modules directly. Atomic species are one-hot encoded over H/C/N/O/F. Two interaction blocks, each followed by three on-site layers, carry 80 features with residual softplus transformations. Every interaction uses 20 learned inverse-distance sensitivities initialized between 1.7 and 10 Bohr and smoothly cut off at 15 Bohr. Orders 0, 1, and 2 are read from the input and the two interaction-block outputs. The implementation has 234,337 parameter values, of which 234,322 are trainable after the least-squares order-zero elemental energies are frozen.

We restore the paper’s Glorot-uniform initialization [Glorot and Bengio, 2010]; the maintained package otherwise defaults to Xavier-normal for several tensors. Biases begin at zero. Higher-order energy heads are initialized at $\sigma_E 10^{-2n}$. For a minibatch D the optimized loss is exactly

$$\mathcal{L}_D = \frac{\text{MAE}_D + \text{RMSE}_D}{\sigma_E} + 10^{-6} \mathcal{L}_{L2} + 10^{-2} \langle R \rangle_D, \quad (3)$$

where the energy-head contribution to $\mathcal{L}_{L2}$ is divided by σ_E^2 as in the original paper. Training is energy-only.

Adam uses learning rate 10^{-3} , $(\beta_1, \beta_2) = (0.9, 0.999)$, and minibatches of 30 [Kingma and Ba, 2015]. The learning rate is fixed for the first 100 epochs. Thereafter, 50 epochs without a new validation-MAE minimum halve it. Training stops after two such halvings without an intervening minimum or at 2,000 epochs, and the minimum-validation checkpoint is restored. Every epoch, learning-rate transition, checkpoint, split, and wall time is retained. Evaluation stores the energy, R , the alternative ratio-of-sums hierarchy score, all $\hat{E}_i^n$, and all six HiPerCal inputs for every nine-heavy-atom molecule.

Published HIP-NN values are kept as external rows, not training targets. The primary anchors are $0.256 \pm 0.003 \text{ kcal mol}^{-1}$ at $N_{\text{train+val}} = 110,426$ and 0.354 ± 0.004 at 50,000 [Lubbers et al., 2018, Table I]. Their random full-QM9 tests are not the nine-heavy-atom OOD test used here. To make the smaller training pool interpretable, we also fit the descriptive power law $\text{MAE} = aN^{-b}$ to all three published Table I HIP-NN points. It predicts $0.501 \text{ kcal mol}^{-1}$ at 21,799 training-plus-validation molecules. Because molecule-size composition also changes, this is a scale check, not a claim of equivalent test distributions.

2.3 HiBreak: discovery and confirmation of chemical failure motifs

Before residual evaluation, we registered sixteen candidates spanning topology, electronic structure, composition, and conformation. RDKit 2026.03.4 parses the GDB SMILES shipped inside each QM9 record [RDKit contributors, 2026]. Graph candidates include small and multiple rings, fused/spiro rings, aromatic and extended- π systems, multiple non-aromatic unsaturations, adjacent heteroatoms, heteroatom-rich compositions, paired carbonyls, nitrile-plus- π systems, multiple fluorines, formal charge separation, flexible chains, and high branching. Coordinates independently define a donor-bound H within 2.5 Å of a nonbonded heteroatom and a nonbonded heavy-atom pair within 2.25 Å. For these two geometry flags only, atoms are treated as bonded when their distance is below 1.25 times the sum of tabulated covalent radii (H/C/N/O/F: 0.31/0.76/0.71/0.66/0.57 Å); the donor and acceptor must be distinct, and a directly bonded donor-acceptor pair is excluded. Appendix A gives every operational definition and result.

On discovery molecules, each candidate is compared with its complement using one-sided Mann-Whitney tests for both ensemble absolute error and ensemble-mean R [Mann and Whitney, 1947]. We report rank-biserial effects. Benjamini-Hochberg correction is applied

across all sixteen candidates separately for error and R [Benjamini and Hochberg, 1995]. A candidate is eligible when it has at least 25 discovery matches, positive effects for both outcomes, and $q_{\text{error}}, q_R \leq 0.05$. HiBreak contains at most six eligible candidates, ordered greedily by the number of previously uncovered discovery errors above 1 kcal mol⁻¹; the product of its two positive effect sizes breaks ties. Selection and order are then frozen.

The final half supplies coverage, conditional error distributions, R , failure attribution, and a second complete set of multiplicity-corrected tests. Seed means carry standard errors across the three trained models. For compactness, after adding the first k frozen motifs we compute

$$A_k = \frac{\#\{m : |\hat{E}_m - E_m| > 1, m \text{ matches one of motifs } 1:k\}}{\#\{m : |\hat{E}_m - E_m| > 1\}}. \quad (4)$$

K_{80} is the smallest k with $A_k \geq 0.8$; no value is assigned if the registered library does not reach that threshold. Per-molecule absolute order sums and the ratios $\sum_i |E_i^1| / \sum_i |E_i^0|$ and $\sum_i |E_i^2| / \sum_i |E_i^1|$ provide mechanism evidence rather than a second selection criterion. The Pareto order is therefore the discovery-frozen greedy order, not an oracle re-sorting on final failures; its K_{80} is an out-of-sample compactness measurement.

2.4 HiPerCal: monotone calibration of a hierarchy profile

HiPerCal predicts the magnitude, not merely the rank, of a molecular energy error. For model replicate s , its registered raw feature vector is

$$\begin{aligned} x_1 &= \log(1 + R_s), & x_2 &= \log\left(1 + \sum_i |\hat{E}_{i,s}^2|\right), \\ x_3 &= \log\left(1 + \frac{\sum_i |\hat{E}_{i,s}^2|}{\sum_i |\hat{E}_{i,s}^1| + 10^{-8}}\right), & x_4 &= \log(1 + \max_i R_{i,s}), \\ x_5 &= \log(1 + \text{sd}_s \hat{E}_s), & x_6 &= \log(1 + \text{sd}_s R_s). \end{aligned} \quad (5)$$

The first four describe failure within the energy hierarchy; the last two measure sample-standard-deviation disagreement among the three fixed-protocol replicas. Ensemble HiPerCal uses the replica mean for the first four features. Seed-resolved fits retain that seed’s internal profile and the same two ensemble-spread features, allowing the reported calibration statistics to carry a between-seed standard error.

Each feature is transformed by its calibration-set empirical CDF. On those transformed features, nonnegative least squares fits $z = \beta_0 + \sum_{j=1}^6 \beta_j x_j$ to absolute error, with every $\beta_j \geq 0$ [Lawson and Hanson, 1995]. Pool-adjacent-violators isotonic regression then learns a nondecreasing $\hat{\epsilon} = g(z)$ in kcal mol⁻¹. This final rescaling follows the general distinction between uncertainty ranking and calibrated regression [Kuleshov et al., 2018]; it is not a probabilistic HIP-NN retraining. An R -only isotonic map fitted to the same molecules is the direct baseline.

Calibration-prediction deciles define ten bin edges, which are applied unchanged to the final half. With bin b , ECE is

$$\text{ECE} = \sum_{b=1}^{10} \frac{N_b}{N} \left| \bar{\hat{\epsilon}}_b - \overline{|\hat{E} - E|}_b \right|; \quad \text{nECE} = \frac{\text{ECE}}{\overline{|\hat{E} - E|}}. \quad (6)$$

The high-error label is consistently $|\hat{E} - E| > 1$ kcal mol⁻¹. Among calibration-score quantiles 0.50, 0.51, ..., 0.99, the threshold maximizing F1 is frozen. We report the full final-set precision-recall sweep, the chosen confusion matrix, and the twenty largest false alarms and misses.

2.5 Off-equilibrium transfer on ANI-1x

ANI-1x supplies roughly five million precomputed $\omega\text{B97X}/6\text{-}31\text{G(d)}$ configurations sampled by active learning, molecular dynamics, normal-mode perturbation, and related procedures

[Smith et al., 2020]. Before model inference, NumPy PCG64 seed 20260731 selects an equal-cap random sample stratified by the 618 eligible formula groups: HCNO only, at most eight heavy and 29 total atoms, at least four available configurations, and at most twenty sampled configurations per formula. Keeping ANI-1x molecules inside the training heavy-atom range isolates conformation and electronic-structure transfer from the nine-heavy-atom size shift used for QM9 diagnosis. The resulting 12,170 configurations are split equally within each formula into offset-fit and evaluation halves.

QM9 and ANI-1x use different density functionals and basis sets, while the HIP-NN target is an atomization energy and the ANI file stores total energy. For formula f and model s , the offset-fit half therefore determines

$$c_{f,s} = \text{median}_{m \in f, \text{fit}} (\hat{E}_{m,s} - E_m^{\text{ANI}}). \quad (7)$$

The disjoint evaluation target is the relative error $|\hat{E}_{m,s} - E_m^{\text{ANI}} - c_{f,s}|$. Formula centering removes a constant composition/method offset but deliberately leaves conformational and isomeric variation. Thus the experiment tests motif transfer, not absolute cross-functional accuracy and not a newly trained ANI potential.

RDKit infers connectivity and bond order directly from each sampled evaluation geometry, with a finite bond-order search cap. Graph assignment succeeds for 6,015 of 6,085 structures; all 70 failures remain in overall energy metrics but are excluded from conditional motif tests. Every frozen HiBreak motif is tested for positive error and R effects, with Benjamini-Hochberg correction over the frozen library. We also report the fraction of formula strata having at least two matching and two nonmatching evaluation structures in which the motif has the same larger-error direction. A one-sided binomial test against probability one half, again BH-corrected over estimable frozen motifs, treats those formula strata rather than configurations as replication units. A fluorine-dependent motif, if selected, is reported as unstable rather than extrapolated to this HCNO dataset.

2.6 Uncertainty and computational accounting

The trained model is the random unit of replication. Unless explicitly marked “ensemble,” model metrics are the mean and standard error over three seeds. Supports and hash-partition counts are deterministic corpus properties and therefore are not assigned a seed error bar. Ensemble statistics are point estimates of a single three-model committee. All Mann-Whitney tests compare molecules, while seed variation is separately shown for effect magnitudes and operational metrics. Inference and cheminformatics timings exclude file serialization and are measured on the staged runtime; training wall time includes validation and checkpoint writes.

3 Results

3.1 A faithful baseline encounters a consequential size shift

Table 1 keeps external and new measurements in separate rows. On the in-domain validation splits, the three replicas reach 0.318 ± 0.011 kcal mol⁻¹ MAE. This is below the published 0.354 ± 0.004 value at 50,000 random train-plus-validation molecules and the 0.501 kcal mol⁻¹ descriptive value obtained by extrapolating the three published learning-curve points to our smaller 21,799-molecule pool. The comparison is a fidelity anchor rather than a leaderboard claim: the published tests are random full-QM9 subsets, whereas our validation molecules contain at most eight heavy atoms.

The sequestered size shift is much harder. Across every nine-heavy-atom molecule, MAE is 0.789 ± 0.015 kcal mol⁻¹, RMSE is 2.060 ± 0.076 kcal mol⁻¹, and $18.72 \pm 0.34\%$ of predictions exceed 1 kcal mol⁻¹; the 99th-percentile absolute error is 7.493 ± 0.137 kcal mol⁻¹. The untouched final half gives 0.787 ± 0.017 kcal mol⁻¹, $18.69 \pm 0.30\%$, and a 7.426 ± 0.209 kcal mol⁻¹ 99th percentile, consistent with the full OOD population rather than an unusually easy analysis split. The original score remains informative—its Spearman association with absolute error is 0.368 ± 0.019 —but the large residual spread at a given R motivates calibration beyond ranking. Averaging the three predictions reduces the full-set point-estimate

Table 1: Two-baseline accuracy and diagnostic results. Published rows are external random-split QM9 measurements from Tables I and II of Lubbers et al. [2018]; their RMSE and tail rate correspond to the 80-feature largest split. New values are mean $\pm$ standard error over three replicas. "This work" uses the same energy predictions as the reimplemented row and adds final-half calibration and the frozen motif library. Dashes mark quantities the source did not report or that do not apply.

Method	evaluation population	train+val	MAE	RMSE	$> 1 \text{ kcal mol}^{-1}$	ECE	compactness
HIP-NN (paper, Table I/II)	random QM9	110,426	0.256 ± 0.003	0.527 ± 0.020	$2.26 \pm 0.07\%$	—	—
HIP-NN (paper, Table I)	random QM9	100,000	0.261 ± 0.002	—	—	—	—
HIP-NN (paper, Table I)	random QM9	50,000	0.354 ± 0.004	—	—	—	—
HIP-NN (reimplemented)	≤ 8 -heavy validation	21,799	0.318 ± 0.011	0.672 ± 0.099	$4.67 \pm 0.52\%$	—	—
HIP-NN (reimplemented)	all 9-heavy OOD	21,799	0.789 ± 0.015	2.060 ± 0.076	$18.72 \pm 0.34\%$	—	—
HiBreak + HiPerCal	untouched 9-heavy final half	21,799	0.787 ± 0.017	2.068 ± 0.088	$18.69 \pm 0.30\%$	0.016 ± 0.003	2

Table 2: The frozen HiBreak library on final evaluation. Support gives count and population coverage. M/N denote matching/nonmatching molecules; seed-level MAEs and failure attribution are mean $\pm$ standard error, while median/P90 uses the three-seed ensemble residual. q_{err} is the final-half Benjamini-Hochberg value across all sixteen registered candidates; all error columns are in kcal mol^{-1} .

Motif	operational definition	support (coverage)	MAE_M	median/P90 _M	MAE_N	failure attribution (%)	q_{err}
close nonbonded heavy	a nonbonded heavy-atom pair is closer than 2.25 Å	32,734 (60.1%)	0.944 ± 0.023	0.336/1.505	0.552 ± 0.008	73.6 ± 0.1	7.4×10^{-245}
proximal donor acceptor	a donor-bound H lies within 2.5 Å of a nonbonded N, O, or F atom	11,793 (21.6%)	0.742 ± 0.009	0.323/1.262	0.800 ± 0.019	23.6 ± 0.3	7.6×10^{-18}
polycyclic	at least two RDKit SSSR rings	29,026 (53.3%)	1.001 ± 0.028	0.369/1.608	0.544 ± 0.003	70.5 ± 0.5	$< 10^{-300}$
extended pi	a connected component of at least four heavy atoms incident to aromatic, double, or triple bonds	13,058 (24.0%)	0.906 ± 0.009	0.346/1.626	0.750 ± 0.021	31.4 ± 0.7	1.3×10^{-87}
small ring	at least one RDKit SSSR ring of size three or four	34,416 (63.1%)	0.829 ± 0.022	0.302/1.237	0.715 ± 0.010	64.2 ± 0.7	2.0×10^{-14}
hetero rich	at least four N, O, or F atoms	9,487 (17.4%)	0.975 ± 0.015	0.422/1.705	0.748 ± 0.017	26.3 ± 0.2	1.4×10^{-210}

MAE to $0.648 \text{ kcal mol}^{-1}$; no retraining or energy correction is introduced by the analyses below.

3.2 HiBreak gives compact failure attribution, but broad coverage

The discovery screen freezes 6 motifs in this order: close nonbonded heavy, proximal donor acceptor, polycyclic, extended pi, small ring, hetero rich. Table 2 then evaluates each one only on the untouched final half. It reports support, a seed-resolved conditional MAE, ensemble median and 90th percentile, the nonmatching MAE, attribution of all $> 1 \text{ kcal mol}^{-1}$ failures, and a fresh sixteen-hypothesis error q value. Thus a broad motif cannot appear persuasive merely because it covers many molecules, and a narrow motif cannot hide its support.

The selected structures do not merely shift error; they also move the hierarchy in the registered direction. Figure 1 places all candidates in joint error/ R effect space and shows the final-half order-decay profiles of the frozen library. The corresponding matching and non-matching R , $|E^1|/|E^0|$, and $|E^2|/|E^1|$ values are reported without aggregation in Table 5. These diagnostics distinguish an early failure to improve beyond dressed atoms from a late failure to suppress the second interaction correction.

The order profiles resolve a different architectural burden for every frozen motif. A close nonbonded heavy pair has mean $R = 0.168$ versus 0.097 for its complement; its median $|E^1|/|E^0|$ and $|E^2|/|E^1|$ ratios rise from 0.0434 to 0.0484 and from 0.0341 to 0.0428, respectively. Such a pair is not assigned a direct covalent bond, yet the distance interaction couples two dressed local environments and the second block must contextualize that contact. Polycyclic molecules show the same late-order signature: $R = 0.176$ versus 0.098, with ratios 0.0475/0.0449 and 0.0435/0.0347 (match/nonmatch). Multiple short closed paths can deliver overlapping context to an atom, consistent with the especially weak second-order decay. Small rings produce a milder version ($R = 0.156/0.113$, order ratios 0.0467/0.0457 and 0.0397/0.0384): short, angular closed paths and strained geometry are not explicit inputs, so the two interaction corrections carry the distinction.

The remaining motifs reveal why a single arithmetic mean of R is insufficient. Proximal donor-acceptor structures have a lower mean R (0.125/0.144) despite a positive final rank-

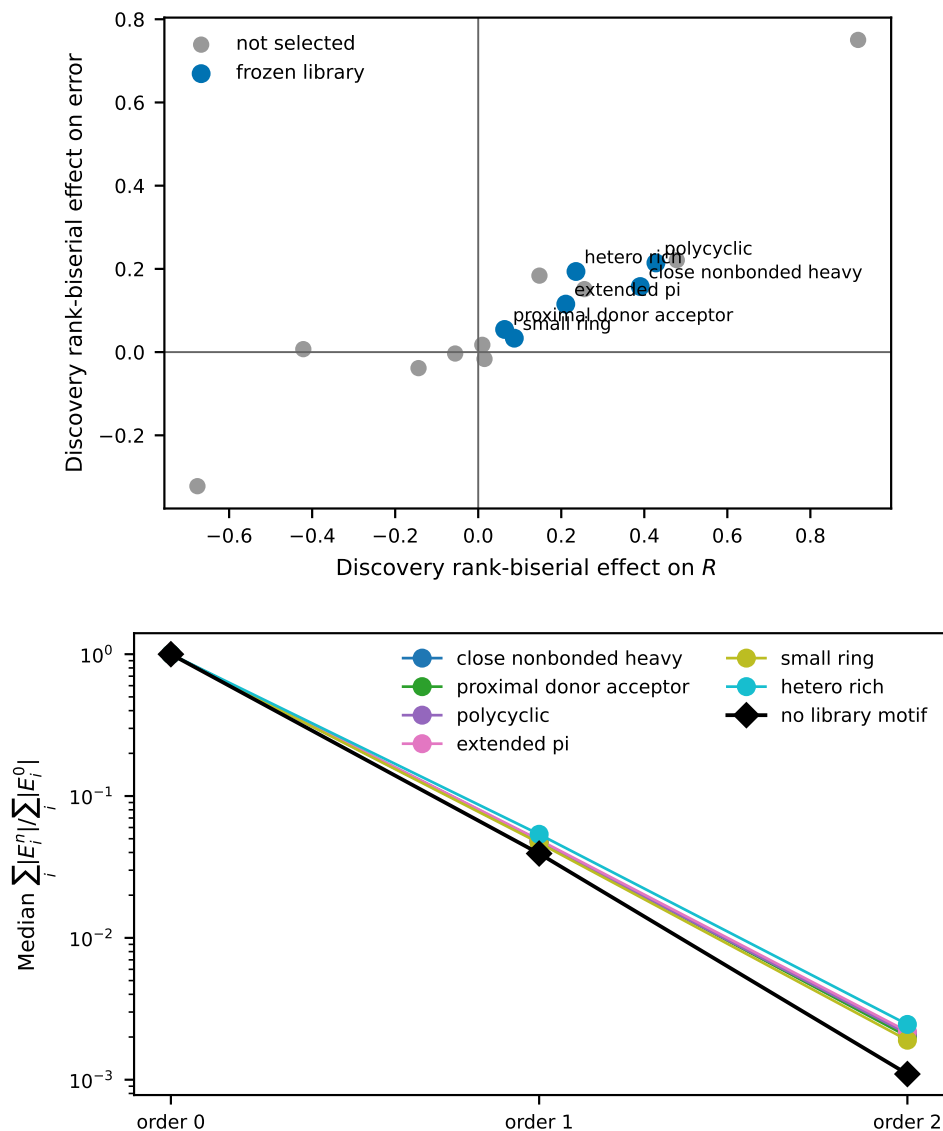


Figure 1: Chemical and hierarchical mechanism diagnostics. Top: discovery rank-biserial effects on ensemble error and R for every registered candidate; blue points form the frozen library. Bottom: final-half median absolute order sums normalized to order zero. A slowly decaying second correction identifies hierarchy strain that a graph label alone cannot establish.

biserial R effect, while both median decay ratios increase (0.0484/0.0458 and 0.0411/0.0386). This distributional shift is consistent with a directional through-space environment whose first correction detects the contact and whose second correction resolves its surrounding donors and acceptors. Extended- π structures likewise have mean $R = 0.137/0.141$ but elevated ratios 0.0491/0.0456 and 0.0441/0.0376; delocalized connectivity extends beyond a single local update and leaves a larger late correction. Finally, hetero-rich molecules have mean $R = 0.136/0.141$ yet the largest order-ratio uplifts, 0.0537/0.0452 and 0.0455/0.0377. A species-only order-zero head cannot distinguish coupled polarization and heterogeneous heteroatom environments, so those molecules rely more heavily on both interaction heads. These are architecture-consistent associations, not assignments of the learned terms to unique physical many-body energies.

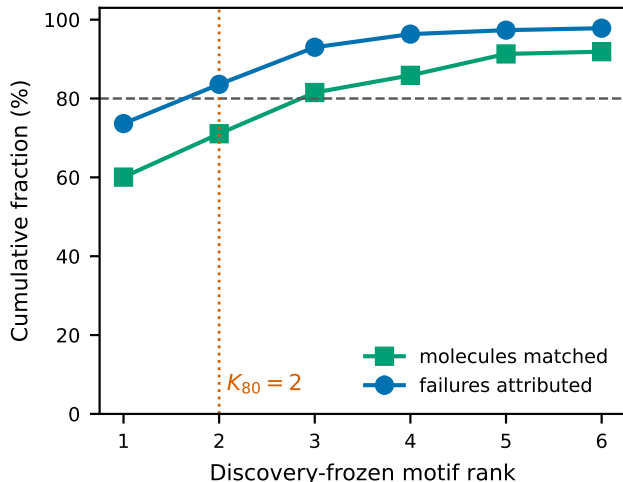


Figure 2: Final-half HiBreak compactness. Motifs retain the discovery-frozen greedy order. Error bars are standard errors over model replicas; the molecule-coverage curve is a deterministic corpus property.

The greedy order is also a compactness test, not a collection of independent anecdotes. On final evaluation, $K_{80} = 2$. At the complete frozen library, $97.8 \pm 0.1\%$ of seed-resolved failures are attributed while 91.9% of all molecules match at least one selected motif. Figure 2 exposes both curves, preventing a high attribution number obtained only by labeling the entire population. Specifically, the first two motifs attribute $83.6 \pm 0.2\%$ of failures but already match 71.1% of molecules. HiBreak is therefore compact in number of concepts, not a sparse alert rule: the six-motif $97.8 \pm 0.1\%$ attribution accompanies 91.9% population coverage. Its value is chemical organization of the tail, whereas selective abstention requires the continuous calibrator below.

3.3 HiPerCal turns a ranking signal into error units

On the untouched final half, HiPerCal obtains ECE 0.016 ± 0.003 kcal mol⁻¹ and normalized ECE 0.020 ± 0.004 . The identically calibrated R -only baseline has ECE 0.013 ± 0.003 kcal mol⁻¹, so the richer profile’s replicated ECE is 16.6% higher rather than lower. HiPerCal’s largest mean bin gap is 0.060 ± 0.020 kcal mol⁻¹; $100.0 \pm 0.0\%$ of nonempty seed-level deciles lie within 10% of their observed mean errors. Every decile, including tied-map empty bins if any, appears in Table 7. The ensemble reliability curve shown in Figure 3 has ECE 0.008 kcal mol⁻¹, compared with 0.013 for its R -only map; these are committee point estimates rather than additional replicas. 100.0% of its nonempty deciles satisfy the same 10% criterion.

The fitted coefficients explain the mixed comparison. Prediction disagreement carries the largest NNLS coefficient in all four fits (0.611 – 1.073), while R , the order-2/order-1 ratio, and disagreement in R are also retained every time. The raw order-2 magnitude is excluded in three of four fits, and maximum atomwise non-hierarchicality is small or excluded. Thus the profile adds reproducible ranking information, especially committee disagreement, but does not uniformly improve mean-error calibration over an isotonic map of R alone. The ensemble point estimate reverses the ECE ordering (0.008 versus 0.013), so neither aggregation level supports an unqualified calibration win.

The calibration-selected operating point yields $49.9 \pm 1.1\%$ precision and $65.9 \pm 1.1\%$ recall across seeds, with a $15.2 \pm 0.5\%$ false-positive rate and $34.1 \pm 1.1\%$ false-negative rate. The matched R -only procedure yields $38.5 \pm 0.6\%$ precision, $56.9 \pm 3.1\%$ recall, $20.9 \pm 1.5\%$ false positives, and $43.1 \pm 3.1\%$ false negatives. For the ensemble point estimate, the frozen predicted-error threshold is 0.802 kcal mol⁻¹, producing 45.4% precision and 59.2% recall, compared with 39.9% and 57.0% for R alone. Figure 3 shows the full reliability range and all

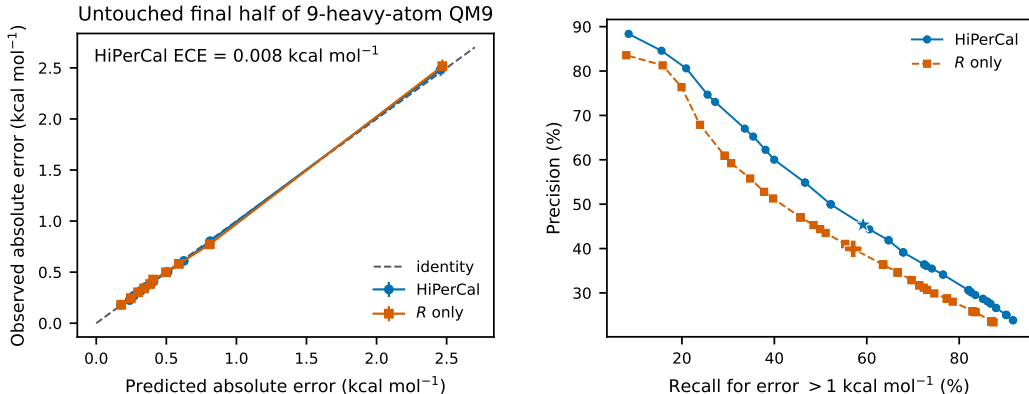


Figure 3: Final-half uncertainty evaluation. Left: predicted versus observed absolute error for HiPerCal and an R -only isotonic baseline; bars are standard errors across molecules and the dashed line is identity. Right: precision-recall values obtained by applying every calibration-quantile threshold unchanged; the star and pentagon mark the respective calibration-F1 operating points.

fifty calibration-frozen thresholds for each calibrator, rather than only the chosen operating points.

3.4 Off-equilibrium transfer is motif dependent

The frozen ANI-1x sample contains 12,170 configurations from 618 formula strata, of which 6,085 form the disjoint evaluation halves. Formula-centered relative-energy MAE is 24.622 ± 0.437 kcal mol $^{-1}$ across replicas and 24.433 kcal mol $^{-1}$ for their mean. Connectivity assignment succeeds for 98.85% of evaluation geometries.

Of 6 HiBreak motifs present in the HCNO sample, 6 retain the positive aggregate error direction and 4 retain positive error and R effects after correction over the frozen family. Independently, 0 pass the BH-corrected formula-stratum direction test. Figure 4 compares effect directions directly; Table 10 adds support and both levels of corrected significance. The mixed result is scientifically useful: it separates failure modes carried by chemical connectivity from those dependent on equilibrium geometry.

The configuration-level successes are close nonbonded heavy pairs (matched/nonmatched mean relative error 26.23/22.13 kcal mol $^{-1}$), proximal donor-acceptor environments (26.08/22.80), polycyclic molecules (33.62/23.36), and hetero-rich compositions (23.98/23.92); all four have BH-supported positive error and R effects. Extended- π systems are unsupported for both outcomes, whereas small rings retain a positive error effect but not a positive R effect. None of the six passes the independent formula-stratum correction; hetero-rich is not estimable within formula because the flag is composition-defined. Consequently the off-equilibrium evidence is configuration-level association, not replicated formula-level universality. The overall formula-centered MAE, 24.622 ± 0.437 kcal mol $^{-1}$ with $95.1 \pm 0.2\%$ above 1 kcal mol $^{-1}$, also makes this a deliberately severe transfer diagnosis rather than an ANI accuracy claim.

4 Relation to uncertainty and chemical interpretability

Modern MLIP uncertainty work has largely asked how to add epistemic or aleatoric information to a point predictor. Committees use disagreement to control generalization and active learning [Schran et al., 2020]; dropout gives an approximate Bayesian route [Wen and Tadmor, 2020]; scalable Bayesian sampling seeks a posterior over network parameters [Thaler et al., 2023]; and latent or feature-space models seek a cheaper single-network proxy [Zhu et al., 2023]. Direct comparisons find no single-model method uniformly superior to

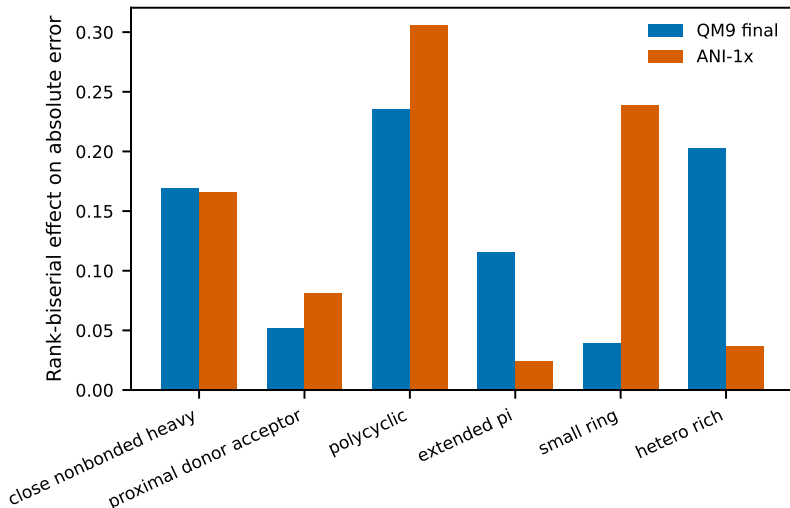


Figure 4: Conformational transfer of the discovery-frozen motif library. Rank-biserial effects compare motif-matched and nonmatched ensemble absolute errors on the QM9 final half and on formula-centered ANI-1x evaluation configurations. No ANI result participates in motif selection.

ensembles [Tan et al., 2023], and recent foundation-model studies continue to find method- and domain-dependent confidence quality [Bilbrey et al., 2025, Wimer et al., 2026].

Calibration is a separate layer of the problem. Ensemble scales can be recalibrated for energy and force distributions [Busk et al., 2023]; conformal prediction provides coverage under its exchangeability conditions [Hu et al., 2022]; and newer conditional maps trade some simplicity for sharper environment-dependent intervals [Ho et al., 2026]. HiPerCal is closest in spirit to post-hoc monotone recalibration, but differs in object and claim. It does not learn a variance head, latent distance, posterior, or conformal interval. It asks whether the three terms already produced by a fixed 2018 hierarchy contain a calibratable signature of that hierarchy’s own truncation failure. The nonnegative profile makes the calibration diagnostically legible, while the R -only fit isolates what is gained beyond the original score.

Atomwise energy decompositions have also been used to recover chemical trends from deep molecular models [Schütt et al., 2017]. HiBreak changes the inferential direction: registered chemical concepts are tested as predictors of hierarchy failure under a controlled size shift, then confirmed on unused molecules and transferred to off-equilibrium geometries. This sample-split, multiplicity-corrected motif attribution is complementary to feature visualization. Its goal is a compact applicability-domain description that a chemist can inspect alongside a numerical error forecast.

5 Scope, failure analysis, and cost

5.1 What remains unexplained

HiBreak is an applicability-domain summary, not an exhaustive partition of error. Even after the full frozen library, $2.2 \pm 0.1\%$ of seed-resolved predictions above 1 kcal mol^{-1} occur in molecules that match no selected motif. Conversely, a motif match is not itself a failure: the conditional distributions in Table 2 overlap their complements. This is expected because motifs overlap one another, broad graph classes contain easy members, and atomization energy can expose interactions that none of sixteen binary descriptors names. The residual suggests that a larger prospective library should add continuous local-geometry and electronic environment descriptors, but doing so would require a new registration and a new confirmatory corpus rather than inspecting the misses here.

HiPerCal has a different blind spot. At the ensemble operating point it produces 5,242 false alarms and 3,000 misses on the final half. The most severe examples in Table 9 show both directions: isotonic plateaus can assign the same predicted magnitude to molecules with very different realized residuals, while hierarchy corrections and committee spread can remain small when all replicas share a bias. HiPerCal estimates a conditional mean absolute error; it does not provide a prediction interval or a finite-sample coverage guarantee. Its flag should therefore trigger a higher-fidelity calculation or abstention, not certify unflagged structures.

The largest miss makes that limitation concrete. QM9 42066, C1CC=CCCC##CC1, has ensemble error 12.310 kcal mol⁻¹ but predicted error 0.681, $R = 0.098$, and no HiBreak match. In the opposite direction, QM9 49110, O=CC1=C2CN3C(C1)C23, has actual error 0.185 kcal mol⁻¹ but lands on the 8.800 kcal mol⁻¹ top isotonic plateau with $R = 2.015$ and four motif matches (close nonbonded, polycyclic, extended π , and small ring). The former is shared confident bias outside the registered chemical vocabulary; the latter shows that hierarchy strain can reflect computational difficulty even when cancellation leaves the total energy accurate.

5.2 Interpretation limits

The atomic terms are learned decompositions, not uniquely defined physical many-body energies. Interaction order expands the message-passing receptive field, so a weakened order-1 or order-2 decay is evidence about the network’s computational burden rather than proof of a particular quantum-mechanical mechanism. Likewise, the final-half tests support association under the specified size shift, not causal effects of functional groups. The registered definitions, dual error/ R screen, and order-ratio diagnostics make the associations falsifiable, but they do not remove correlations among ring topology, unsaturation, heteroatom content, and compact geometry. Moreover, R is an atomwise sum. Fixing the test population at nine heavy atoms removes its largest size confounder, but hydrogen count still varies; the intensive order ratios and maximum atomwise signal help distinguish decay changes from a pure count effect without eliminating that limitation.

The corpus also bounds the claim. QM9 contains small neutral HCNOF molecules at minimized geometries, and training is energy-only. Charged systems, radicals, reactions, condensed phases, forces, and elements beyond fluorine are untested. ANI-1x widens the geometry distribution but restricts chemistry to HCNO and changes the electronic-structure level. Formula centering makes its relative-energy comparison meaningful, yet formula groups contain both conformers and constitutional isomers, and graph perception fails for 70 highly distorted configurations. Those structures remain in the global error summary but cannot support motif-conditional conclusions. Moreover, QM9 motifs use the archived GDB graph whereas ANI-1x graphs are inferred from coordinates; transfer therefore includes a representation shift as well as a geometry shift. The in-domain fidelity number is also the 1,000-molecule set used for stopping, not a second untouched small-molecule test. It checks implementation scale under the prescribed split, but the comparison with the paper’s random-QM9 test remains descriptive; all new confirmatory claims instead use unreused nine-heavy-atom molecules. Finally, all calibration evidence comes from a fixed partition of the same nine-heavy-atom population; a deployment shift beyond that population calls for recalibration. Three independent replicas meet the study’s replication criterion and expose seed variability, but their standard errors should not be read as high-precision estimates of optimizer-induced variation. The replicas also share most of the 21,799-molecule pool, differing in initialization, minibatch order, and the 1,000-molecule validation draw; committee spread is therefore not a complete measure of dataset uncertainty.

5.3 Computational tradeoff

Each paper-parity replica trains for 1.433 ± 0.245 hours on one NVIDIA H200, stopping between epochs 659–1245 with best checkpoints between epochs 559–1145. A single checkpoint evaluates a nine-heavy-atom molecule, including all atomic hierarchy terms, in 0.020 ± 0.000 ms. A three-member committee therefore spends approximately three forward passes; HiPerCal adds no retraining. Its four seed/ensemble fits, the four matched R -only fits, and all held-out predictions amortize to 0.0058 ms per OOD molecule. Registered motif assign-

ment costs 0.029 ms per molecule on twelve CPU workers. On ANI-1x, inference costs 0.021 ± 0.012 ms per configuration and bond perception costs 0.058 ms.

The quality-cost choice is consequently explicit. R and the atomic order profile are effectively free once a HIP-NN forward pass is required; committee spread receives the largest fitted weight and improves high-error triage at threefold inference cost, but it does not improve replicated ECE over R alone; RDKit motifs add a small CPU-side diagnosis. A latency-constrained use can fit the same monotone procedure to one seed, whose complete calibration records are reported, while a high-stakes screening workflow can retain all three replicas and use the reported precision-recall sweep to choose a more conservative threshold.

6 Conclusion

HIP-NN’s hierarchy is diagnostically useful, but only after separating three questions that a rank correlation cannot answer. Under a strict heavy-atom-count shift, HiBreak finds a small, chemically named set of structures whose errors and hierarchy scores both increase on unreused molecules; the frozen library attributes $97.8 \pm 0.1\%$ of seed-resolved failures. HiPerCal then converts the full hierarchy profile into absolute-error units. It does not beat the identically fitted R -only map on replicated ECE (0.016 ± 0.003 versus 0.013 ± 0.003 kcal mol⁻¹), although the ensemble point estimate does; instead, its clear gain is high-error triage, reaching $49.9 \pm 1.1\%$ precision and $65.9 \pm 1.1\%$ recall versus $38.5 \pm 0.6\%$ and $56.9 \pm 3.1\%$. The remaining confident mistakes show that this is a conditional-mean forecast, not a safety certificate.

The ANI-1x experiment supplies the necessary boundary on the chemical claim: 4 of 6 testable motifs retain joint, multiplicity-corrected error and hierarchy support off equilibrium, but none passes the independent formula-stratum direction test. Thus some failure signatures persist as configuration-level associations across geometry distributions, whereas stronger formula-level transfer is unresolved and others are tied to minimized QM9 structures. The practical outcome is not a universal warning light. It is an auditable applicability-domain layer—a predicted error magnitude, a chemical reason, and an explicit record of where both can fail—built without altering the underlying energy model.

Reproducibility statement

All training, evaluation, molecule-level hierarchy outputs, chemical assignments, calibration bins, threshold sweeps, checkpoints, and seed-resolved summaries accompany this paper. The QM9 split rule and analysis registrations predate checkpoint evaluation; the ANI-1x sample, graph assignments, and formula-stratum amendment predate ANI model inference. The artifact manifest gives SHA-256 digests, and the integrity audit verifies the heavy-atom separation, cross-seed alignment, energy reconstruction from atomic hierarchy terms, and the rendered paper.

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Table 4: Complete motif screen. r_{err} and r_R are rank-biserial effects (match versus non-match); q is Benjamini–Hochberg adjusted. Eligibility precedes greedy failure coverage and therefore can include motifs not selected among the first six.

Candidate	n_{disc}	r_{err}	q_{err}	r_R	q_R	Eligible	final r_{err}	final q_{err}
small ring	17,058	0.033	3.1×10^{-6}	0.087	2.8×10^{-33}	yes	0.039	2.0×10^{-14}
polycyclic	14,509	0.214	3.1×10^{-204}	0.428	$< 10^{-300}$	yes	0.235	$< 10^{-300}$
fused or spiro	13,386	0.221	1.5×10^{-217}	0.478	$< 10^{-300}$	yes	0.240	$< 10^{-300}$
aromatic ring	4,648	0.151	2.3×10^{-59}	0.256	4.6×10^{-166}	yes	0.157	1.6×10^{-123}
extended pi	6,540	0.115	5.0×10^{-45}	0.211	5.6×10^{-146}	yes	0.115	1.3×10^{-87}
multiple unsaturation	6,808	0.017	0.022	0.010	0.148	no	0.007	0.173
hetero adjacent	2,532	0.184	1.7×10^{-52}	0.147	2.0×10^{-34}	yes	0.186	2.7×10^{-104}
hetero rich	4,848	0.194	3.0×10^{-99}	0.235	5.6×10^{-146}	yes	0.202	1.4×10^{-210}
carbonyl pair	1,228	-0.003	0.714	-0.056	1.000	no	-0.006	0.847
nitrile with pi	1,971	-0.039	1.000	-0.144	1.000	no	-0.044	1.000
polyfluoro	141	0.007	0.589	-0.421	1.000	no	0.026	0.308
formal charge separation	150	0.750	1.3×10^{-56}	0.915	2.1×10^{-83}	yes	0.696	3.3×10^{-77}
flexible chain	2,652	-0.322	1.000	-0.676	1.000	no	-0.308	1.000
high branching	11,642	-0.016	1.000	0.015	0.024	no	-0.020	1.000
proximal donor acceptor	5,954	0.054	1.1×10^{-10}	0.064	4.9×10^{-14}	yes	0.052	7.6×10^{-18}
close nonbonded heavy	16,298	0.158	2.2×10^{-107}	0.390	$< 10^{-300}$	yes	0.169	7.4×10^{-245}

A Registered candidate library

Table 3 gives all candidate definitions exactly as applied. Table 4 retains every hypothesis, including those that did not enter HiBreak. Discovery q values correct sixteen tests separately for error and R ; final q values are a new sixteen-test correction on the untouched final half.

Table 3: Residual-independent candidate definitions registered before model evaluation.

Candidate	Operational definition
small ring	at least one RDKit SSSR ring of size three or four
polycyclic	at least two RDKit SSSR rings
fused or spiro	two SSSR rings share at least one atom
aromatic ring	at least one fully aromatic RDKit ring
extended pi	a connected component of at least four heavy atoms incident to aromatic, double, or triple bonds
multiple unsaturation	at least two non-aromatic double or triple bonds
hetero adjacent	a direct N-N, N-O, or O-O bond
hetero rich	at least four N, O, or F atoms
carbonyl pair	at least two C=O substructures
nitrile with pi	a C#N substructure plus at least one additional multiple or aromatic bond
polyfluoro	at least two fluorine atoms
formal charge separation	both positive and negative formal charges in the parsed molecular graph
flexible chain	at least three Lipinski rotatable bonds under RDKit’s strict definition
high branching	a heavy atom with at least four heavy-atom neighbors
proximal donor acceptor	a donor-bound H lies within 2.5 Angstrom of a nonbonded N, O, or F atom
close nonbonded heavy	a nonbonded heavy-atom pair is closer than 2.25 Angstrom

B Seed-resolved optimization and accuracy

The three epoch logs contain the initial evaluation (epoch zero), every optimizer epoch, all validation values, and each learning-rate transition. Intermediate checkpoints are written

Table 5: Hierarchy-decay diagnostics on the untouched final half. R is a three-seed mean with standard error. The two ratios use the median three-seed ensemble absolute order sums for matching (M) and nonmatching (N) molecules; q_R is Benjamini–Hochberg corrected over all sixteen registered candidates.

Motif	R_M	R_N	q_R	$ E^1 / E^0 _M$	$ E^1 / E^0 _N$	$ E^2 / E^1 _M$	$ E^2 / E^1 _N$
close nonbonded heavy	0.168 ± 0.015	0.097 ± 0.002	$< 10^{-300}$	0.0484	0.0434	0.0428	0.0341
proximal donor acceptor	0.125 ± 0.005	0.144 ± 0.009	2.5×10^{-21}	0.0484	0.0458	0.0411	0.0386
polycyclic	0.176 ± 0.017	0.098 ± 0.002	$< 10^{-300}$	0.0475	0.0449	0.0435	0.0347
extended pi	0.137 ± 0.002	0.141 ± 0.010	3.0×10^{-268}	0.0491	0.0456	0.0441	0.0376
small ring	0.156 ± 0.014	0.113 ± 0.002	4.8×10^{-68}	0.0467	0.0457	0.0397	0.0384
hetero rich	0.136 ± 0.006	0.141 ± 0.009	2.6×10^{-277}	0.0537	0.0452	0.0455	0.0377

Table 6: Every formal training replicate. “OOD all” uses all 109,024 nine-heavy-atom molecules; “OOD final” uses the hash-defined confirmatory half. Wall time includes validation and checkpoint serialization.

Seed	best epoch	validation MAE	OOD-all MAE	OOD-final MAE	OOD > 1 (%)	hours
117	559	0.333	0.819	0.820	19.34	0.945
223	1088	0.326	0.771	0.766	18.18	1.630
337	1145	0.296	0.778	0.776	18.65	1.723

every 100 epochs in addition to `best.pt` and `last.pt`. The split files retain source-array indices and QM9 identifiers, allowing the heavy-atom separation to be audited without rerunning training.

C Complete calibration bins and operating failures

Table 7: All ensemble HiPerCal deciles on the final half. Edges and the map were fitted on the disjoint calibration quarter. SD, median, and P90 describe the full within-bin absolute-error distribution.

Bin	n	predicted MAE	observed MAE	SD	median	P90	absolute gap
1	5,298	0.177	0.182	0.173	0.136	0.392	0.004
2	3,297	0.238	0.226	0.214	0.170	0.476	0.012
3	7,434	0.267	0.268	0.248	0.202	0.573	0.001
4	5,301	0.316	0.315	0.302	0.237	0.678	0.001
5	4,916	0.353	0.356	0.358	0.262	0.756	0.003
6	6,408	0.409	0.410	0.431	0.295	0.884	0.001
7	5,365	0.509	0.501	0.569	0.350	1.079	0.008
8	5,115	0.625	0.610	0.715	0.420	1.312	0.015
9	5,116	0.813	0.803	1.065	0.539	1.683	0.009
10	6,250	2.458	2.482	4.679	1.143	5.651	0.024

The fifty thresholds for each ensemble calibrator, from calibration quantiles 0.50 through 0.99, and their final confusion matrices are retained in `logs/threshold_sweep.csv`. The corresponding HiPerCal and R -only sweeps for every seed and the ensemble are in `logs/threshold_sweep_seed_resolved.csv`. The ensemble bin records used here are in `logs/calibration_bins.csv`; the companion `logs/calibration_bins_seed_resolved.csv` retains both calibrators for all four fits, including any empty bin created by tied isotonic predictions. No ensemble bin was suppressed in Figure 3.

Table 8: Nonnegative least-squares coefficients before isotonic remapping. Signals are the empirical-CDF transformed quantities in Eq. 5; a zero is an exact NNLS exclusion.

Model	intercept	R	$\sum_i E_i^2 $	$ E^2 / E^1 $	$\max_i R_i$	prediction SD	R SD
117	0.0000	0.3239	0.0000	0.1731	0.0295	1.0495	0.3114
223	0.0000	0.2858	0.0148	0.1717	0.0000	1.0174	0.2659
337	0.0000	0.1895	0.0000	0.1891	0.0192	1.0734	0.3069
ensemble	0.0000	0.2509	0.0000	0.3642	0.0908	0.6112	0.1748

Table 9: Largest operating-point errors on the final half. A miss has actual error above 1 kcalmol⁻¹ but a score below the calibration-selected threshold; a false alarm is the converse. The complete top-twenty lists, including SMILES, are in logs/calibration_failure_cases.csv.

Type	QM9 ID	actual	predicted	R	HiBreak matches
miss	42066	12.310	0.681	0.098	none
miss	42110	9.529	0.536	0.148	extended pi
miss	38731	9.033	0.681	0.119	polycyclic, small ring
miss	42108	8.896	0.754	0.127	extended pi
miss	123523	8.360	0.536	0.141	close nonbonded heavy, polycyclic, extended pi, hetero rich
miss	37122	8.193	0.551	0.461	close nonbonded heavy, polycyclic, small ring
false alarm	49110	0.185	8.800	2.015	close nonbonded heavy, polycyclic, extended pi, small ring
false alarm	34484	0.191	8.800	1.902	close nonbonded heavy, polycyclic, extended pi, small ring
false alarm	41178	0.439	8.800	2.378	close nonbonded heavy, polycyclic, small ring
false alarm	36998	0.574	8.800	1.758	close nonbonded heavy, polycyclic, small ring
false alarm	72712	0.576	8.800	1.799	close nonbonded heavy, polycyclic, small ring
false alarm	41165	0.608	8.800	2.203	close nonbonded heavy, polycyclic, small ring

D ANI-1x motif transfer

Table 10: Per-motif off-equilibrium transfer. Effect is the ANI-1x configuration-level rank-biserial error effect; q_{cfg} corrects the frozen, testable HiBreak family. The stratum rate is the percentage of eligible formula strata with larger matched error, and q_{str} is the BH-corrected one-sided binomial value for a rate above one half.

Motif	support	error effect	q_{cfg}	stratum rate	q_{str}	outcome
close nonbonded heavy	2,663	0.166	6.1×10^{-28}	53.7%	0.167	configuration
proximal donor acceptor	2,105	0.081	1.5×10^{-7}	55.6%	0.097	configuration
polycyclic	344	0.305	1.7×10^{-21}	63.8%	0.074	configuration
extended pi	1,992	0.024	0.064	39.3%	0.991	directional/unsupported
small ring	602	0.238	1.1×10^{-21}	54.5%	0.227	directional/unsupported
hetero rich	2,471	0.037	0.009	not estimable	—	configuration

The sampling record lists all 618 formula strata, available configuration counts, sampled indices, and fit/evaluation allocation. The per-configuration archive retains formula, source index, inferred SMILES where available, all motif flags, three predictions, three hierarchy scores, and three formula-centered errors. No electronic-structure calculation was run.

E Artifact-to-claim map

Corpus and split	logs/qm9_preparation.json, logs/split_seed_*.npz, and logs/motif_assignment_summary.json.
Optimization	logs/hyperparameters_seed_*.json, logs/training_seed_*.csv, and logs/training_summary_seed_*.json.
Per-molecule model output	logs/test_outputs_seed_*.npz and logs/validation_outputs_seed_*.npz.
Motif inference	logs/motif_candidate_discovery.csv, logs/motif_candidate_final.csv, logs/motif_library_final.csv, and logs/motif_compactness.csv.
Calibration	logs/calibration_models.json, logs/uncertainty_outputs.npz, logs/calibration_bins_seed_resolved.csv, and logs/threshold_sweep_seed_resolved.csv.
Conformational transfer	logs/ani1x_sampling.json, logs/ani1x_motif_assignment_summary.json, and logs/ani1x_transfer_summary.json.
External baseline	logs/paper_reference.md and logs/paper_reference_extrapolation.json.